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Overconfidence in Interval Estimates

Jack B. Soll
INSEAD

Joshua Klayman
University of Chicago

Judges were asked to make numerical estimates (e.g., "In what year was the first flight of a hot air balloon?"). Judges provided high and low estimates such that they were X% sure that the correct answer lay between them. They exhibited substantial overconfidence: The correct answer fell inside their intervals much less than X% of the time. This contrasts with choices between 2 possible answers to a question, which showed much less overconfidence. The authors show that overconfidence in interval estimates can result from variability in setting interval widths. However, the main cause is that subjective intervals are systematically too narrow given the accuracy of one's information—sometimes only 40% as large as necessary to be well calibrated. The degree of overconfidence varies greatly depending on how intervals are elicited. There are also substantial differences among domains and between male and female judges. The authors discuss the possible psychological mechanisms underlying this pattern of findings.

When people say they are X% sure about a fact, they are right less than X% of the time. This finding, generally known as overconfidence, has been observed in many studies over the past several decades. However, the veracity and generality of the phenomenon have been increasingly challenged in recent years. The magnitude of overconfidence depends on how the questions are chosen, what they are about, and how confidence judgments are elicited, and in some cases underconfidence is observed (Juslin, Wennerholm, & Olsson, 1999; Klayman, Soll, González-Vallejo, & Barlas, 1999). Moreover, when overconfidence occurs, it is not clear what the underlying causes are. Early research attributed overconfidence to cognitive biases such as anchoring and insufficient adjustment, biased retrieval of information, and confirmation bias in interpreting the information retrieved (e.g., Hoch, 1985; Koriat, Lichtenstein, & Fischoff, 1980). More recent work has pointed instead to the effects of unbiased imperfections and variations in information and in judgments. Researchers have shown that noisy but unbiased judgments can produce systematically biased outcomes under many circumstances (Erev, Wallsten, & Budescu, 1994; Juslin & Olsson, 1997; Pfeifer, 1994; Soll, 1996). This seems to be a sufficient explanation for many cases of overconfidence, although the extent to which noise accounts for overconfidence remains controversial (Ayton & McClelland, 1997; Brenner, 2000; Wallsten, Erev, & Budescu, 2000).

In this article, we examine the effects of bias and noise in a type judgment that often exhibits extreme overconfidence, namely *interval estimates.* In these, participants provide estimates of a quantity in terms of ranges or boundaries that correspond to a given degree of confidence. An example is as follows: "When was Charles Dickens born?" "I'm 80% sure the answer is between 1750 and 1860." This contrasts with the two-choice format used in most studies of subjective confidence. In a typical design, participants are given questions with a choice of two answers, one of which is correct. For each question, the judge chooses an answer and then states his or her confidence, on a scale from 50% to 100%, that the choice is correct: "Who was born first, Charles Darwin or Charles Dickens?" "Dickens—80% sure."

Although less studied than two-choice questions, interval estimates are analogous to many real-world judgments. When people decide when to leave for the airport, or how much to invest in stocks, or how much inventory to hold, they implicitly make judgments about a plausible interval for the time the ride will take, how much they will have in their account 20 years from now, or the rates of sales and production. Yaniv and Foster (1995, 1997) found an analogous process in ordinary verbal communications that they called *grain size.* People imply a rough sense of confidence in an interval estimate by choosing the precision with which they express information. "I think it was during the last half of the 19th century" implies a different degree of confidence than "I think it was around 1875." Listeners expect speakers to choose a grain size appropriate to their level of knowledge. Interval estimates are also widely used in the practice of decision analysis, and overconfidence can have serious consequences for the accuracy of decisions predicated on those estimates (Clemen, 1996; Spetzler & Staël von Holstein, 1975; von Winterfeldt & Edwards, 1986).

There is little doubt that overconfidence predominates in interval judgments. For example, judges' 90% intervals typically contain the correct answer less than 50% of the time (e.g., Klayman et al., 1999; Russo & Schoemaker, 1992; see also Alpert & Raiffa, 1982; Lichtenstein, Fischhoff, & Phillips, 1982). Moreover, interval estimates have displayed substantially more overconfidence than have two-choice questions in direct comparisons using the same stimuli (Juslin et al., 1999; Klayman et al., 1999). However,

Jack B. Soll, INSEAD, Fontainebleau, France; Joshua Klayman, Center for Decision Research, Graduate School of Business, University of Chicago.

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Correspondence concerning this article should be addressed to Joshua Klayman, Center for Decision Research, Graduate School of Business, University of Chicago, 1101 East 58th Street, Chicago, IL 60637. E-mail: jack.soll@insead.edu or joshk@uchicago.edu

little progress has been made toward understanding the processes, cognitive or stochastic, that underlie interval estimates. It has been suggested that interval estimates involve an initial attempt at a "best guess" point estimate, plus insufficient adjustment from that point in defining lower and upper bounds (Tversky & Kahneman, 1974). It has also been proposed that interval estimates are more prone to confirmation biases than are two-choice questions because there are no clearly specified alternatives (Klayman et al., 1999).

Our approach to investigating the processes that underlie confidence has two main components. First, we present a model for understanding how variability in judgments affects interval estimates, and we describe methods for estimating those effects. Existing models of the effects of variability on overconfidence have been developed only for two-choice questions. As we explain later, those models do not apply to interval estimates. It is interesting to note though, that variability can translate into overconfident interval estimates in a different way. We introduce new methods for analyzing interval estimates that allow us to estimate how much overconfidence is attributable to the biasing effect of variability, and how much is attributable to overly narrow intervals, variability aside.

Second, we examine the contributions of variability and narrow intervals to overconfidence in three studies of interval estimates. The studies used three different elicitation procedures, all of which set a target level of confidence and ask the judge to describe an interval that corresponds to that confidence. In the *range* method, the judge was asked for an 80% interval, as in the Dickens question above. The *two-point* method breaks the task into two separate questions: "I am 90% sure that Dickens was born after ____" and "I am 90% sure than Dickens was born before ____." The *three-point* method consists of the two-point method plus an additional question: "I think that it is equally likely that Dickens was born before or after the year ____."

We found that all three formats produce overconfident judgments, but that they do so to different degrees. Variability contributes to overconfidence, but variability alone cannot account for the degree of overconfidence observed or for differences among formats. The overall level of overconfidence suggests an important role for biases and limitations in information processing, in contrast to current views about two-choice questions. Comparing the three formats can provide additional clues as to the nature of the information processing involved.

In comparing the formats, we consider the accuracy of judges' estimates (e.g., how close the center of their interval is to the correct answer) as well as the width of the interval provided (cf. McKenzie & Amin, 2002). Appropriate confidence consists of choosing interval widths appropriate to the accuracy of one's estimates. Why would our three elicitation formats differ? Consider first the contrast between the range and two-point formats. Judges could well treat these as equivalent. However, there are reasons to believe that the two formats differ psychologically. We hypothesized that a range estimate is treated as a single (fuzzy) judgment, even though one must use two numbers to describe it. Thus, people conduct a single, limited search of the information available to them and estimate the plausible range suggested by that search. In contrast, we hypothesized that the two-point method encourages people to sample their knowledge twice, once for a low estimate and again for a high estimate. Sampling two subsets of knowledge rather than one should lead to more accurate estimates. This can be thought of as analogous to the improvement in accuracy that is obtained by averaging the opinions of two judges who have nonredundant sources of information (Hogarth, 1978; Soll, 1999). At the same time, two-point estimates may be less prone to cognitive bias than are ranges. Drawing different implications from two sets of evidence may be easier than drawing different implications from a single set of evidence. This would lead to wider intervals in the two-point format. Indeed, we found that two-point estimates show less overconfidence than do range estimates because of both greater accuracy and wider intervals.

What about three-point estimates? Requiring explicit attention to a best-guess-like point estimate should exacerbate any tendency to anchor on a best guess, making intervals more narrow (Russo & Schoemaker, 1992). On the other hand, some studies (Block & Harper, 1991; Clemen, 2001; Juslin et al., 1999; Selvidge, 1980) have found that asking for a best guess in conjunction with intervals reduces overconfidence. Asking for three estimates instead of two may lead judges to retrieve more evidence. This may lead to more accurate judgments of where to center one's interval or to perceptions of greater variability and uncertainty in evidence (Block & Harper, 1991; Juslin et al., 1999). Our results tend to support the last hypothesis: Three-point estimates show the least overconfidence, and increased interval width seems to be an important component of that.

Variability and Overconfidence

Since about 1990, the traditional emphasis on cognitive biases in confidence has largely been replaced by explanations based on the effects of variability. These explanations decompose subjective probability assessments into judgment and error. The former may be thought of as the true judgment (Erev et al., 1994) or as the mean response over replications (Brenner, 2000). The latter refers to error in the statistical sense: nonsystematic variations that have no net overall bias. We prefer to call this *variability* to more clearly distinguish it from other possible meanings of *error*. There are several sources of variability, ranging from having accidentally had atypical experiences (Juslin, 1994; Soll, 1996), to variability in feelings of confidence under identical conditions (Erev et al., 1994), to unreliability in generating a number that goes with one's feeling of confidence (Budescu, Erev, & Wallsten, 1997; Juslin, Olsson, & Björkman, 1997). The proportion of correct answers is also susceptible to variation depending on which particular questions happen to be sampled. Given that both confidence and accuracy are noisy measures of something else (the validity of the underlying information), mean-reversion effects are inevitable. That is, high confidence is usually too high and low confidence is usually too low—the typical pattern of miscalibration. Similarly, when accuracy is low for a set of questions, it is likely to be lower than the available information would lead one to expect, and when accuracy is high, it is likely to be higher than one would expect (Dawes & Mulford, 1996; Juslin, Winman, & Olsson, 2000; Klayman et al., 1999; Suantak, Bolger, & Ferrell, 1996). This leads to overconfidence for "hard" question sets, and occasionally even underconfidence for "easy" sets—the so-called hard–easy effect.

Models of variability have lent considerable insight into the two-choice task, so it is natural to explore ways that variability might affect interval estimates as well. Juslin et al. (1999) sug-

gested one such model. They extrapolate from the two-choice task by postulating that, for example, a 90% subjective interval can be modeled as two separate two-choice questions: one eliciting an estimate of 95% certainty about the lower limit and one eliciting an estimate of 95% certainty about the upper limit. They assert that the effect of noise is to make each of these estimates overly extreme. A typical finding is that expressions of 95% certainty are associated with roughly 80% accuracy. Thus one might expect that a 90% interval would miss about 20% at each end and thus would include only 60% of answers.

The explanation proposed by Juslin et al. (1999) appears to explain much of the overconfidence observed in interval estimates. In fact, though, the primary explanation for miscalibration in two-choice questions does not apply to the interval task. With two-choice questions, judges are given fixed alternatives, and they respond with a range of different degrees of confidence. If the researcher then examines different subsets of responses, the mean-reverting effect discussed earlier applies: The higher the expressed confidence, the more likely it is to include some positive error, in other words, to be overconfident. So 95% confidence judgments, being very high, are inevitably too high on average. With interval estimates, the confidence level is fixed, and the judge must determine the answer that fits it. The researcher considers all judgments, not a subset. Thus, mean-reversion effects are not expected.

A Model of Variability in Interval Estimates

Although mean reversion cannot explain miscalibration in interval estimates, there is another subtle way that variability can lead to overconfidence. To explain how this can happen, we present three mythical figures: a minor goddess by the name of Calibra, a sage named Serge, and a peasant named Peggy.

Calibra. Calibra, being only a minor goddess, is not omniscient. However, unlike some gods, she is honest, and she is also perfectly calibrated. Calibra's knowledge gives her a sense of how close any given value is likely to be to the truth. If you ask her the height of Mount Olympus, for example, her knowledge indicates that 300 m is very likely to be too small, 1,000 m is probably still too small, 7,000 m is almost surely too big, and so on. Because she is a goddess, Calibra's sense of likelihoods corresponds precisely to a well-calibrated subjective probability density function (SPDF). When she says that she is C% certain that the answer is $A \pm D$, she is right C% of the time. You can check this yourself. Over her indefinite lifetime, Calibra has made myriad judgments of this type and has kept precise records, and the correct answer is always revealed and recorded subsequent to her judgments. You can draw a random subset of judgments from the records at any level of confidence and check her hits and misses. You will find that the samples behave exactly as though they were drawn by randomly sampling a series of independent, random events of probability C.

Serge. Serge is a very sage sage. His subjective probability judgments are like those of the goddess Calibra, except that being mortal, he knows less and is prone to error. Specifically, when asked for an interval corresponding to C% confidence, he gives the same interval that perfectly calibrated Calibra would give if she had Serge's knowledge, plus some noise. That is, Serge reports an interval of $A \pm (D + e)$, where e has a mean of zero and does not vary with A or D.

Peggy. Peggy is a peasant. She is not as wise as Serge, but she is no dummy. Her confidence judgments do correlate positively with her chance of being correct. Like Serge, there is noise in her judgments, but her intervals are also systematically narrower than they ought to be. With the same knowledge as Serge, Peggy reports an interval of $A \pm (bD + e)$, where b is a constant between zero and one. We call this latter problem *narrowing.*

Calibra is perfectly calibrated, and Peggy is overconfident on average. But what can one expect from Serge? As it turns out, Serge's noise will also produce overconfidence, even though it is itself unbiased. This makes it difficult to know the extent to which Peggy's overconfidence is due to bias beyond the effects of variability.

Why Serge is overconfident. Serge's noise translates into overconfidence because of a characteristic of the SPDF implied by his knowledge. The most likely single value (the mode of the SPDF) is inside the interval, and the likelihood goes down moving away from that peak. This seems likely to be true of very many SPDFs. This implies that variability in setting the endpoints of the intervals has asymmetric effects depending on whether it produces inward errors (making the interval narrower) or outward errors (making the interval wider).

Consider the normal distribution shown in Figure 1. Suppose that we ask Serge to estimate the year in which Charles Darwin was born. Figure 1 shows the curve that Calibra would have if she knew exactly what Serge knew. Calibra would say, for example, that there is an 80% chance that Darwin was born between 1800 and 1840 (Interval I, shown by the solid, vertical lines). Noisy Serge, though, might set an interval that is 10 years too narrow. So, he says he is 80% sure the year is between 1805 and 1835 (Interval J). Calibra would have said there was 66% chance that the answer lies in that interval, and she is perfectly calibrated. Over a set of such questions, Serge's inward errors produce overconfidence.

But Serge is not biased. He is equally likely to make the opposite error, making his interval 10 years too wide (Interval K). Calibra would tell us that the probability that the answer lies between 1795 and 1845 is not 80%, as Serge reports, but 89%.

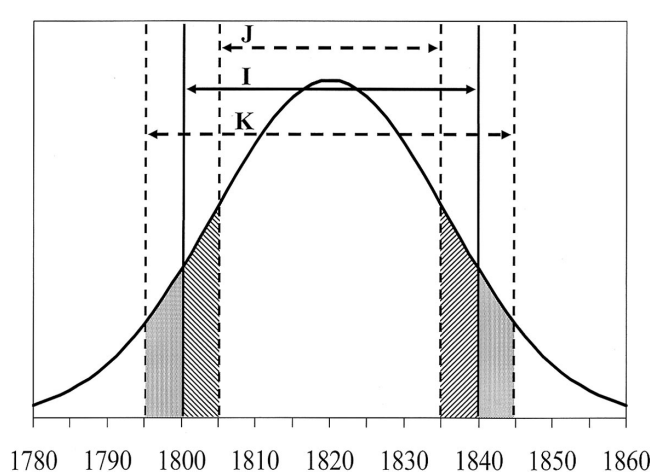

Figure 1. A hypothetical subjective probability density function for an estimate of the year in which Charles Darwin was born. Intervals J and K represent opposite 10-year errors in estimating the interval, I, that contains 80% of the probability.

Over a set of questions, Serge's outward errors would produce underconfidence. Note, though, that the area mistakenly excluded by an inward error (the striped area) is greater than the area mistakenly added by an outward error (the gray area). Specifically, a 10-year error produces 14% overconfidence in one direction and 9% underconfidence in the other. On balance, then, variation in interval width produces overconfidence, even if intervals are "the right size" on average. This example uses a normal curve, but this property is true of any curve that slopes upward toward a peak within the interval and downward on the other side.

In other words, Serge's unbiased noise on the response scale translates into a bias in probabilities, because of the shape of the probability function. Whether we call Serge biased depends on the scale of evaluation. He is unbiased on the response scale but overconfident nonetheless. Indeed, in Serge's case, overconfidence is entirely due to variability and not bias.

Why is Peggy overconfident? When we check the records on Peggy, we find that she is considerably more overconfident than Serge is. This could be because Peggy's intervals are narrower than the intervals Calibra and Serge would provide with the same knowledge. However, it could also be that Peggy's judgments are noisier than Serge's. In the above example, if Peggy's errors in each direction were 30 years instead of Serge's 10, they would produce 55% overconfidence and 18% underconfidence, and Peggy would be quite overconfident on average. This raises the question of how to tell the extent to which an overconfident person is suffering from the effects of variability, narrowed intervals, or both.

Measuring the Effects of Variability

We use three methods for estimating what the right interval size is for judges like Peggy. Each method involves some simplifying assumptions and approximations. However, we hope that the different methods converge on plausible estimates.

Compare the Inferred and the Observed Accuracy of Estimates

In this approach we pretend that each judge has a particular SPDF for each question. We doubt that judges carry probability density functions in their heads, but as long as they have a sense that some answers are more probable than others, we can model judges as though they have SPDFs. We can estimate their SPDFs from the fractiles they give us. For example, if Serge tells us he is 90% sure that Darwin was born after 1800 and 90% sure he was born before 1840, and we assume a normal SPDF, we can infer the curve shown in Figure 1, which has a median (and mean) of 1820 and a standard deviation of 15.6.

The SPDF implied by fractile estimates in turn implies a distribution of expected surprises. There should be a 20% chance that the correct answer is more than 20 years away from 1820 (10% that it is earlier than 1800 + 10% that it is later than 1840), a 52% chance that it is more than 10 years away, a 75% chance that it is more than 5 years away, and so on. Summing across all such probabilities for the curve shown in Figure 1, we find an expected absolute deviation (EAD) from the median of 12.4 years. Of course, this estimate of the EAD is based on the assumption of a normal SPDF, but it turns out that it serves as a good approximation for a wide array of functions, including asymmetrical and uniform functions.[1]

We can then look at a given judge's answers across a set of questions and compare two measures of surprise. The first is the observed mean absolute deviation (MAD) between the median of his or her SPDF and the true answer. The second is the mean of the EADs implied by each pair of fractiles he or she gives (the MEAD). Fortunately for Calibra, she is endowed with a second kind of good calibration. She has appropriate expectations about her accuracy; that is, her MEAD equals her MAD (except for some random variation due to luck of the draw in sampling questions). It is important to note that the same is true for Serge, even though he is overconfident. As we noted before, given Serge's SPDF, a 10-year mistake in one direction causes 14% overconfidence, whereas the same error in the other direction causes 9% underconfidence. However, because EAD is a linear function of the interval width, equal and opposite mistakes in reporting interval width have equal and opposite effects on EAD (in this case, changing from 12.4 years to 10.8 or to 14.0 years, respectively). So even though Serge's intervals are sometimes too narrow and sometimes too wide, on average his MEAD will equal his MAD. As for Peggy, if her MEAD is less than her MAD, that means she is receiving bigger surprises than she expects, and her intervals are too narrow.

The ratio M = MEAD/MAD represents the ratio of actual to ideal interval size, disregarding the effects of variability. To explain, let us again assume normal SPDFs. For any normal distribution, EAD $= W/(z_{.5+(p/2)}\sqrt{2\pi})$, where W is the width of the interval, p is the probability contained in the interval, and z is the value of the standard normal distribution. So if we average across all of Peggy's intervals, MEAD $= \overline{W}/(z_{.5+(p/2)}\sqrt{2\pi})$. Now, if Calibra were in Peggy's shoes, she would be perfectly calibrated, and her average interval width would be just right so that her MEAD equalled her MAD. Call that well-calibrated average width $\overline{W^*}$, where MAD $= \overline{W^*}/(z_{.5+(p/2)}\sqrt{2\pi})$. Combining the preceding two equations, MEAD/MAD $= \overline{W}/\overline{W^*}$. Thus, M represents the ratio of observed average interval width to the well-calibrated, zero-variability interval width. If Peggy's M is less than one, it indicates that her average interval width is narrower than is justified given the information she is using. This is true whether or not she is susceptible to variability in setting interval widths. In other words, $M < 1$ indicates a bias that cannot be attributed to random error or any other source of variability.

Calibra, of course, has an M of 1, as does Serge, because his intervals are on average the same width as Calibra's would be, given his knowledge. However, Serge is still overconfident be-

[1] For example, the EAD for a uniform distribution on the range 0–1 is .250. On such a distribution, .1 and .9 are the 10th and 90th fractiles. If we mistakenly assume a normal distribution with those fractiles, we would infer that EAD is .249. We have found that the normal distribution provides similarly accurate estimates of EAD for both asymmetric and symmetric beta distributions, as long as they are unimodal or flat. Taking the three fractiles provided for each item in Experiment 2, we used the software package @Risk to infer the best fitting beta function for each item and then computed the expected absolute deviation from the median for that beta (for formula, see Johnson, Kotz, & Balakrishnan, 1995). EAD estimates based on the inferred betas were nearly identical to those inferred from only the high and low estimates and an assumption of normality.

cause of noise. To be well calibrated on probabilities, Serge needs an M sufficiently larger than 1 to compensate for the biasing effects of variability. He cannot simultaneously be unbiased in terms of M and in terms of the probability of catching the correct answer.

Measure the Correlation Between Interval Size and Hits

More variability and narrower intervals both cause more overconfidence. However, the former has an additional effect that the latter does not. The greater the variability, the greater the correlation between the size of the interval and whether the correct answer falls in it (a "hit"). Calibra's intervals vary in size according to how much she knows about a particular question. However, her smallest 80% intervals contain the answer 80% of the time, as do her largest 80% intervals. Serge, on the other hand, also widens and narrows his intervals by accident. When he makes an inward error, his interval narrows and is also less likely to catch the right answer. The opposite happens when he makes an outward error. Thus, for Serge there is a correlation between interval size and hits. For Peggy, we can identify the combination of variability and narrowing that would produce the combination of overall hit rate and size-to-hits correlation that we see in her responses.

How hit rate and correlation map onto variability and narrowing depends on the way that variability operates. We do not actually know how variability operates, so we use two different, plausible models for it. One model we use is

$$W_i/W_i^* = x, \tag{1}$$

where W_i is the width of the interval the judge gives for estimate i, and W_i^* is the well-calibrated size Calibra would have given. The distribution of variable x represents the judge's variability in setting the interval size. In our simulations, we use a gamma distribution for x because it is bounded at 0, and, logically, intervals cannot be less than 0 in size. The other model we use is

$$\log(W_i/W_i^*) = x. \tag{2}$$

With logarithms, interval widths cannot be less than 0, so we can just assume a normal distribution for x. Further details are provided in the Appendix.

In the studies presented here, we find very high correlations (between .85 and 1.0) among our three methods of estimating each participant's average bias, $\bar{x}$. We present analyses using the M measure and note the few instances in which either of the simulation models produce qualitatively different results.

Overview of Experiments

We present three experiments in which we varied how interval estimates were elicited. In each experiment, we assessed the degree of overconfidence, and, using the methods developed in the previous section, Measuring the Effects of Variability, we estimated the extent to which overconfidence is attributable to variability or to narrowing. In Experiment 1, we compared range and two-point estimates in a task that provided a direct comparison to statistically appropriate interval widths (using a kind of statistical Calibra). In Experiment 2, we introduced the three-point format. This served two purposes: to test whether this format shows qualitatively different results and to provide a test of some of the assumptions made in Experiment 1. Specifically, in Experiment 1, we elicited only the two endpoints of the intervals. In our methods for estimating the effect of variability in this study, we assumed that the SPDFs behind those endpoints were normal. With the three fractiles elicited in Experiment 2, we fit beta distributions of a wide variety of shapes, demonstrating that in most cases the normality assumption has little impact on estimates of the effects of variability and bias. Armed with this reassurance, plus results suggesting that there is less overconfidence with three-point than with two-point estimates, we proceeded to compare those two formats directly in Experiment 3. In addition to checking the differences between formats, we checked for gender differences, just to be thorough. Prior studies have shown mixed results concerning possible gender differences in overconfidence (Barber & Odean, 2001). Our studies reveal that male judges are more overconfident than are female judges, but we had no a priori hypotheses about this variable.

Experiment 1

In Experiment 1, we had three main goals. One goal was to see whether two-point estimates produce less overconfidence than do range estimates, consistent with hypotheses about limited or biased information processing. Another important goal was to determine, for both formats, the extent to which overconfidence is attributable to overly narrow intervals versus variability in interval widths. The third goal was to further understand any difference observed between the two formats. Most studies of confidence ask judges to draw information only from their own knowledge and memory. We used a variation introduced by Soll (1996) in which participants make predictions based on objective cue values provided in the study, relying on their knowledge to determine how to combine them into a judgment. For example, a judge might be asked to estimate the price of an automobile, given that it had 6 cylinders, had been rated 4 out of 5 in road tests by *Consumer Reports* magazine, and was medium in size. This permits us to compare participants' accuracy and subjective intervals with objective standards, namely the confidence intervals on predictions for each item provided by a regression model using the same cues.

Method

Participants. Participants were 32 undergraduate and graduate students from the University of Chicago; there were 9 men and 7 women in each of two conditions. They were recruited via notices posted around campus and were paid $9 for the approximately 45 min it took to complete the procedure. Three participants in each of the two experimental conditions were replaced because their estimates revealed that they had misunderstood the questions in one or more of the domains.

Procedure. After signing a consent form, participants were handed 1½ pages of printed task instructions, which they were permitted to keep with them during the procedure. The instructions included the following example:

> Child number 7, selected randomly from the files of the local primary school, weighs 86 pounds, is 51 inches tall, and has 11 permanent (adult) teeth. We . . . then ask you to make an estimate about how old Child 7 might be. We don't ask for a "best guess."

The instructions continued with a description of one of the two elicitation methods:

> Instead, we ask you to give us two numbers such that you are 80% sure that the correct answer lies somewhere between the two.

or

> Instead, we ask that you give us a lower estimate such that you are 90% sure that the child is not younger than that. Then we ask you for an upper estimate such that you are 90% sure that the child is not older than that.

This was followed by an additional paragraph of explanation using the Child 7 example, including a translation of the required probabilities in terms of frequencies ("you would expect on average to miss the answer on about 10 out of the 50 questions" or "you would expect the correct answer to be lower than your lower estimate about 5 times and more than your upper estimate about 5 times"). The remainder of the instructions described how to locate and provide information using the computer.

The estimation task was presented using PCs. Participants worked separately in a room that accommodated up to 4 participants at a time, separated by partitions that prevented them from seeing one another's screens. The task consisted of 50 questions: 12 from each of two domains and 13 from each of two others. The opening screen introduced the first domain. For example,

> Hello! The next 12 questions will ask you to estimate the invoice price of different sedan-type automobiles (that is, the price the manufacturer charges the dealer.) We have randomly selected 12 out of the 67 sedans reviewed in Consumer Reports. Though we do not tell you the makes of the sedans, we give you some statistics about the year 2000 models, along with the low, high, and average for all 67 reviewed sedans. This information was drawn from the statistics available on the Consumer Reports website. These may not necessarily be the most useful statistics for making your estimates. Please hit "PAGE DOWN" to begin.

On each of the next 12 pages, an estimate in that domain was requested. Figure 2 provides an example. Gray boxes represented missing information. The cell marked "Dictionary" provided definitions for the criterion and each of the cues and also the mean and range of the criterion value (i.e., in Figure 2, for the full list of sedans from *Consumer Reports*).[2] Figure 2 shows a screen from the range condition. Participants used the keyboard to enter numbers in each of the two outlined cells in the "between ____ and ____" line. The display in the two-point condition was the same, except that instead of completing the sentence "I'm 80% sure . . . ," there were two response lines that said, "I'm 90% sure that the invoice price of this car is at least [box]" and "I'm 90% sure that the invoice price of this car is at most [box]." Participants could change their responses prior to going on to the next question but could not return to previously entered estimates.

After that, participants were presented with the next domain via the same procedure. After completing 50 questions from four domains, participants were asked, "Out of all 50 questions you answered, for how many of the 50 questions do you think the correct answer will turn out to be within the interval you gave?" They were also asked to provide some demographic information.

Design. Participants were assigned to one of two conditions. Those in the *range* condition were asked for an 80% interval, and those in the *two-point* condition were asked for high and low 90% estimates as described in the *Procedures* section. We used four domains of estimates, and each participant received all four. The order of domains was counterbalanced across participants in a Latin square design. The order of questions within domain was randomized at the beginning of the experiment and was the same for all participants.

There were five possible cues available for each domain, shown in Table 1. Although source materials provided more than five potential cues, the five were chosen to be (a) easily comprehensible and (b) approximately representative of the distribution of cue validities among all the cues contained in the source. On any given trial, three of the five cues were presented. The selection of cues was such that each cue appeared five or six times in the 12 or 13 questions, in different combinations. This meant that different questions within a domain had different quality information, and thus should merit different interval sizes. The assignment of which cues were given on which questions was constant across participants. In each case, the values on the cues were the actual values of a particular item in the source list. Thus, the distribution of cue values and the intercorrelations among cues approximated the population.

Because of a programming error, 2 of the 13 questions in the domain of college quality scores were eliminated from analyses; some participants saw all five cues for those items.

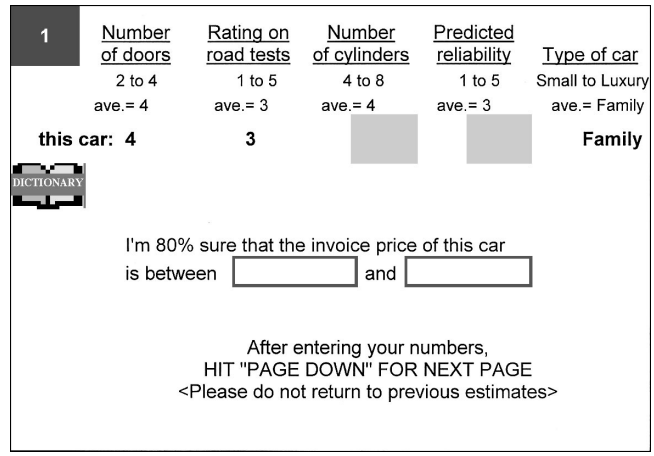

Figure 2. Example of the display used to request estimates in the range-format condition of Experiment 1.

Results

Except where noted, results were analyzed using a multivariate analysis of variance with format (range or two-point) and gender as between-participants variables. For each participant, the dependent measures were sums or averages across the 12 or 13 trials in each domain, producing four measures per participant.[3]

Hit rate. Overall, judges' 80% intervals contained the correct answer 48% of the time. Intervals in the two-point format contained the correct answers more often than they did in the range format, $F(1, 28) = 7.1$, $p = .01$, and there were significant differences among domains $F(3, 26) = 5.9$, $p < .01$, (see Table 2). Female judges had a higher hit rate, $F(1, 28) = 6.5$, $p = .02$. There were no significant interactions.

[2] The mean and range information was provided to reduce participants' feeling of ignorance in using what might be unfamiliar cues and criteria. A study by Klayman and Burt (1998) indicates that provision of this kind of information may reduce overall overconfidence slightly, but not dramatically.

[3] In this and the subsequent experiments, an arcsine transformation was used in analyses of hit rates, and a log transformation was used for interval widths, MAD, and *M*. These are the transformations generally recommended for proportions and for measures that are bounded at zero and thus positively skewed, respectively (Kirk, 1995; Winer, 1971). We report the untransformed values in descriptive statistics.

Table 1
Domains of Estimates and the Cues Provided for Each in Experiment 1

Domain	Available cues
This college's overall quality score	Graduation rate Academic reputation rating % students from top 10% of their high school Classes with over 50 students % faculty who are full time
The average box office gross of '90s movies starring this actress	Number of times she has been nominated for an Oscar Whether she uses her real name Number of films she has starred in Her age Number of children she has
Invoice price of this sedan-type car	*Consumer Reports* predicted reliability *Consumer Reports* rating on road tests Number of doors Size of car Number of cylinders
Winning % of the National Basketball Association team	Average turnovers per game Average assists per game Average rebounds per game Average blocks per game Average points per game

MEAD and MAD. Next, we calculated M = MEAD/MAD as an estimate of how the average size of subjective intervals compares with the average size of interval that would produce good calibration in the absence of noise. For each estimate, we calculated the absolute deviation between the midpoint of the participant's interval and the correct answer. We then averaged these to estimate the MAD for each domain for each participant. We also calculated the MEAD that would result if the participant's announced endpoints were the 10th and 90th percentiles of a normal distribution.

As shown in Table 2, M is substantially less than 1, indicating intervals that are smaller than would be appropriate even in the absence of noise. M is higher in the two-point format than in the range format, $F(1, 28) = 7.0$, $p = .01$. Also, M varied by domain, $F(3, 26) = 6.4$, $p < .01$, and men were more biased than were women, $F(1, 28) = 8.1$, $p < .01$. There were again no significant interactions. Similar results were obtained with both of the simulation-based estimates of narrowing (see Table 2).

It is important to note that M was closer to 1 in the two-point condition for two reasons: MEAD was larger, $F(1, 28) = 5.7$, $p = .02$, and MAD was smaller, $F(1, 28) = 5.6$, $p = .03$.[4] In other words, with the two-point format, intervals were wider but at the same time better centered on the truth. A different pattern underlies the gender difference. Men and women did not differ significantly on MAD ($F < 1$), but they did differ significantly on MEAD, $F(1, 28) = 8.9$, $p < .01$. In other words, the midpoints of the intervals of men and women were equally close to the truth on average, but women had wider intervals.

Comparison to statistical estimates. Another way to gauge participants' confidence is to compare them to statistical estimates. We created linear regression models using each combination of cues seen by participants. These models were created using the whole population from which items were drawn (e.g., all 67 sedans reviewed by *Consumer Reports*, not just the 12 we asked about.) For each item participants saw, we obtained the statistical estimate from the relevant regression model, as well as that model's 80% confidence interval for that observation. As we expected, the models' estimates were more accurate. Participants' MADs were 1.6 times those of the corresponding model's, but their intervals were only .77 times as wide. Participants were both less accurate and more confident than statistical models using the same input data.

Separating out the effect of variability. The fact that M is less than 1 implies that variability alone cannot account for the overconfidence observed. However, variability may still contribute to overconfidence. To estimate the effect of variability, we estimated what each participant's hit rate would have been if his or her intervals had been biased by a constant amount, without variation. Taking each participant's M for each domain, we checked what probability is included in a standard normal distribution between fractiles at $Mz_{.10}$ and $Mz_{.90}$. (Results are similar using average bias estimates from the simulations.) We then compared participants' hit rates to this theoretical, constant standard. The difference is an estimate of the contribution of variability around the average bias. Such variability can contribute to overall overconfidence at higher levels of aggregation, too. The reasoning is analogous to our explanation of the effect of item-by-item variation. The overconfidence of a set of individuals in a given domain may be increased by the net effect of variation from individual to individual, because

[4] We wanted to include all domains in a single analysis, even though their scales were very different (e.g., millions of dollars, percentages). To roughly equate the scales, we standardized MEAD and MAD for each domain using the variance of criterion values found among all items in the source list.

Table 2
Hit Rates and Estimates of Interval Size Relative to Ideal "Calibra Intervals" in Experiment 1

Independent variable	Hit rate[a]	Interval size relative to ideal		
		M	Gamma simulation	Log simulation
Format				
Range	.39 (.22)	.46 (.27)	.46 (.29)	.44 (.29)
Two point	.57 (.20)	.66 (.27)	.71 (.34)	.69 (.34)
Domain				
Movie grosses	.39 (.25)	.44 (.29)	.46 (.39)	.43 (.38)
Basketball wins	.46 (.29)	.57 (.35)	.55 (.43)	.54 (.44)
Car prices	.50 (.29)	.59 (.32)	.63 (.44)	.61 (.44)
College ratings	.57 (.24)	.63 (.31)	.70 (.37)	.68 (.38)
Gender				
Women	.58 (.23)	.70 (.30)	.74 (.36)	.72 (.37)
Men	.40 (.19)	.45 (.22)	.47 (.27)	.45 (.27)

Note. Numbers in parentheses are standard deviations.
[a] Answers that exactly matched an endpoint were counted as correct.

individuals whose intervals are too narrow will have more effect on the overall hit rate than will individuals whose intervals are too wide by the same amount. Similarly, overconfidence across a set of domains may be increased by variations from domain to domain. The effects of each of these sources can be separated, but we found that the total contribution of all three sources of variability to overconfidence was .05 in the range condition and .04 in the two-point condition. Thus, participants were overconfident mainly because their intervals were too narrow given their level of accuracy. Variability in interval width only produces a small amount of overconfidence, and differences in variability cannot account for the large difference between the range and two-point methods.

Posttask estimates of hit rate. After completing all 50 trials, participants were asked to estimate how many of their intervals actually did contain the correct answer. Given that we asked for 80% probability intervals, participants should have been aiming for 40 hits. On average, they estimated 31 hits, and in reality, they had 24. Participants were still overconfident in retrospect, but less so than they were item by item. This finding is less extreme than that hypothesized by Gigerenzer, Hoffrage, and Kleinbölting (1991), who suggested that retrospective estimates concerning a randomly chosen set of items should often show underconfidence. However, our results are in line with those of Sniezek and Buckley (1991), who asked a similar retrospective question for subjective intervals. In any case, this provides further support for the conclusion that interval estimates are prone to fundamental biases to which two-choice questions are not prone. It is also interesting to note that there were positive correlations between retrospective estimates and the actual number of hits of .49 in the range condition ($p = .05$) and .61 in the two-point condition ($p = .01$). (These correlations are not significantly different from each other.) Thus, participants seem to have had some insight after the fact into their degree of overconfidence.

Discussion

The results of this study confirm that interval estimates are prone to substantially more overconfidence than are two-choice questions. In the range format, the hit rate was 39% for 80% intervals, that is, 41% overconfidence. This is similar to the level of overconfidence observed in previous studies (e.g., Juslin et al., 1999; Klayman et al., 1999; Lichtenstein et al., 1982; Russo & Schoemaker, 1992). With two-point estimates, the degree of overconfidence was lower, but it was still substantial (23%). We observed this degree of overconfidence in a task that provided judges with more information than they had had in previous studies (objective cue values and the range and mean of each of the cues and criteria). This presumably reduced judges' reliance on retrieval of information from memory, so it appears that there is plenty of room for bias in other processes as well, such as interpretation of the meaning and importance of the cues. In any case, if the additional information increased judges' accuracy, it increased their confidence as well.

Given that we sampled randomly from all available items in each domain, overconfidence cannot be attributed to the effects of biased sampling of items (cf. Gigerenzer et al., 1991). However, overconfidence could result merely from judges' failure to recognize the biasing effects of variability in interval size. So we used several methods to estimate specifically how the size of subjective intervals compares to what Calibra (i.e., a well-calibrated and noise-free judge) would say. We found that subjective intervals are indeed too narrow—on average, less than 60% of the size they need to be, effects of variability aside. Variability in interval width accounts for only a small amount of overconfidence.

These results confirm our primary hypotheses that interval estimates are seriously biased toward overconfidence, and that this results from overly narrow intervals. Beyond that, we observed interesting differences along three other dimensions: the way in which intervals were elicited, the domain from which items were drawn, and the gender of the participant.

The observed difference between range and two-point formats is intriguing, in that the two ways of asking for intervals appear on the surface to be nearly equivalent. When judges indicated they were 80% sure the answer was between ____ and ____, they were overconfident by 41% and had intervals less than half of the Calibra size. When they separately indicated high and low two-point estimates about which they were 90% sure, they were over-

confident by 23% and had intervals about two-thirds of the Calibra size. We investigated the effects of format differences again in Experiments 2 and 3, so we postpone discussion of this effect until later in this article.

The large differences between domains mirror similar findings in both interval estimates and two-choice questions, showing that different domains of questions are systematically associated with different degrees of overconfidence or underconfidence (Juslin et al., 1999; Klayman et al., 1999; Koehler, Brenner, & Griffin, 2002). The explanation for large, systematic domain differences is still elusive (but see Brenner, 2003). With further investigation, differences between domains could provide valuable information about the sources of bias in confidence. Meanwhile, our findings highlight the risks of relying on any single domain, as a number of studies have done.

Men were almost twice as overconfident as women were in our study. Women's estimates were not more accurate, but their intervals were more than 50% wider than men's. Past studies on gender difference in confidence have produced mixed results. Some studies have found no differences (Lichtenstein & Fischhoff, 1981) and others have reported that men are more overconfident (Pallier, 2003), particularly in "masculine" domains (Barber & Odean, 2001; Lundeberg, Fox, & Punćochaŕ, 1994). Although two of our domains (automobiles and basketball) might be considered stereotypically masculine, we found no interactions between gender and domain on any of our dependent variables. We had no prior hypotheses concerning gender differences in interval judgments, but given the magnitude of the observed effect, it seems worthy of future investigation.

We also find that people realize in retrospect that their hit rate is less than they indicate in individual judgments. Moreover, we find that posttask estimates are correlated with participants' actual hit rates. This suggests that people access different types of evidence for concurrent and retrospective assessments of confidence (Gigerenzer et al., 1991; Griffin & Tversky, 1992; Sniezek & Buckley, 1991). In answering a specific question, people focus on the available information and reasoning for a specific estimate. In contrast, when thinking back over a whole set of estimates, people likely rely largely on a more general sense of the precision of their knowledge in a given domain (Sniezek & Buckley, 1991). Our results suggest that there is some bias at both levels of judgment: Overall, posttask judgments are overconfident, but they are less so than item-by-item judgments.

Experiment 2

In Experiment 1, we used three different methods to obtain estimates of bias in interval width, and they all produced similar conclusions. However, these estimates were necessarily based on only two data each, namely the upper and lower bounds provided by the participant. So we had to assume a lot. We assumed that the judge's median estimate was midway between the two boundaries, and we assumed that participants in the range condition, like the two-point participants, intended to exclude 10% of the probability on each end. A normal distribution is likely to be a good approximation if both assumptions hold true, but perhaps not otherwise.

In the present study, we obtained additional information from judges by asking them to provide explicitly their median estimate as well as separate low and high boundaries. This permits us to take into account the possibility that judges' intervals are asymmetric (see O'Connor & Lawrence, 1989). Having three fractile points instead of two permits us to better approximate the underlying SPDF and to better estimate implied and actual deviations. We used beta functions for this purpose, because they can approximate a great variety of skewed distributions.

As we discovered in Experiment 1, different elicitation formats can produce quite different results. Thus, our attempts to observe behavior more precisely might well affect the behavior observed. As we discussed earlier, prior research offers mixed suggestions about the possible effect of adding a third, median estimate. In fact, results from Experiment 2 suggest that adding the third estimate reduces overconfidence. We test this directly in Experiment 3.

Method

Participants. Participants were 30 master of business administration students at INSEAD's campus in Fontainebleau, France. They were solicited in a commons area, and they received 50 francs (U.S. $7) and a lottery ticket for a task that took roughly 20 min. Two randomly chosen lottery winners received a bottle of Dom Perignon champagne. We did not ask for gender or other demographic information.

Procedures. The experiment was conducted in English because this is the language of instruction at INSEAD. For this study, we returned to the common practice of providing participants with named items and asking them to make estimates on the basis of only their prior knowledge. We used five domains. Three of these domains were traditional "almanac questions," that is, general knowledge questions on arbitrarily chosen topics. We selected domains for which participants would have some knowledge base to draw on and for which there were comprehensive lists of items from which to sample. These were the human fertility rates of different countries, the year in which a variety of devices and processes were invented or discovered, and the average daily high July temperature of major cities around the world.

In addition, we included two domains for which these participants could draw on direct, personal experience and knowledge. These were the enrollments in various courses at INSEAD, and the time required to walk from one place to another around Fontainebleau at a moderate (5 km/hr) rate without interruption.

After reading one page of printed instructions and having an opportunity to ask questions, each participant received a printed booklet of 30 items from one of the five domains, with one item on each page. For each item, they were asked to make three estimates, corresponding to the 10th, 50th, and 90th fractiles of an SPDF. For example, to guess the year in which the telegraph was invented, participants completed the following three statements:

> I am 90% sure that the year is after ____.
>
> I think the year is as likely to be after this year as before it: ____.
>
> I am 90% sure that the year is before ____.

They were permitted to take as much or as little time as they wished, but they were told that people typically finish in about 20 min.

Design. Participants were assigned, alternately, to each of the five domains. All participants in a given domain saw the same 30 items in a different randomized order. The items for each domain were selected at random from the corresponding source list. The order of the three estimate questions varied, with each possible order assigned to 1 participant in each domain. For each participant, the ordering of the three questions remained constant across items.

Results

Except where noted, results were analyzed using an analysis of variance (ANOVA) with domain (five levels), position of the median (first, second, or third estimate) and order of the 10th and 90th fractile estimates (10th before 90th or vice versa) as between-participant variables. Because there was only one participant in each cell, the three-way interaction was not analyzed.

Hit rate. As in Experiment 1, judges were significantly overconfident. Their 80% intervals contained the correct answer 66% of the time. However, the 14% overconfidence observed here is considerably lower than the 41% and 23% found in the range and two-point conditions of Experiment 1. As in previous studies, overconfidence varied across domains, $F(4, 8) = 3.5$, $p = .06$ (see Table 3). There was a marginal effect of the order of 10th and 90th fractiles, $F(1, 8) = 3.5$, $p = .06$, qualified by an interaction between the position of the median and the 10th–90th order, $F(2, 8) = 3.7$, $p = .09$. No other effects approached significance. The order effects appear to have been produced by a single discrepant condition. Participants who gave their 10th fractile, then 50th, then 90th had only 39% hits. In contrast, the hit rates in the other five cells ranged from 60% to 83%. As it turns out, we failed to replicate this order effect in Experiment 3, so this may well have been a chance occurrence.

MEAD and MAD. As in Experiment 1, the ratio *M* provided an estimate of the appropriateness of interval size. Our having three fractile estimates from participants instead of two permitted two refinements to the method for estimating *M* used in Experiment 1. First, we could base our estimates of MEAD on more flexible representations of participants' SPDFs, including asymmetric ones. Second, we could measure MAD using the median estimate provided by the participant rather than assuming the median is in the middle of the distribution. We use M_3 to refer to *M* estimated in this way, from three fractiles, and M_2 to refer to estimates based on only the two endpoints, as in Experiment 1.

Table 3 shows both M_3 and M_2. Note that the two measures agreed well, except in one domain (fertility rates). The two methods produced nearly identical estimates of MEAD (see Footnote 2). However, in the fertility domain, there was a noticeable difference in estimates of MAD. Why did the measures diverge on this one domain? In the fertility domain, 94% of SPDFs were asymmetric, compared to 54% across the other domains. Moreover, the distributions tended to be skewed in a particular way that had a large effect on estimates: The correct answers tended to lie below the midpoint of the interval, whereas the long tail of the SPDF was on the high end.

For Experiment 2, we report our analyses using M_3, because it is based on more information. M_3 differed across domains, $F(4, 8) = 4.9$, $p = .03$. Although no other effects were significant, the pattern of means roughly tracks that for hit rates. This follows from the fact that across participants, M_3 was highly correlated with hit rate ($r = .80$). On average, $M_3 = .96$, which appears much closer to unity than do the values of .46 and .66 for the range and two-point formats of Experiment 1. Results using the estimates of interval bias from simulations are very similar to those using M_3, as shown in Table 3.

Effect of variability. We used the same procedure for estimating the effect of variability that we used in Experiment 1. We found that, of the total observed overconfidence of .14, .11 was attributable to variability. Of that .11, .07 was accounted for by variability across domains. As shown in Table 3, the *M*s ranged from .55 to 1.59. Even though the average *M* across domains was close to 1, the overconfidence produced by domains with $M < 1$ was greater than the underconfidence produced by domains with $M > 1$.

Discussion

Experiment 2 provides information on both methodological and empirical points. The main methodological point concerns how *M* is estimated. When using range or two-point formats, only M_2 estimates are possible. Therefore it is important to have evidence that those estimates, which assume normality of SPDFs, provide a good approximation of *M*. This seems to be the case, except in particular domains in which (a) intervals are heavily skewed and (b) correct answers tend to be consistently toward the thick end or consistently toward the tail of the skewed distributions. In Experiment 3 we used two domains that did not show those characteristics in the present experiment. What about Experiment 1? We looked to see if any of the domains in that study showed a strong tendency of the correct answers to fall on one side of the interval, one of the necessary conditions for M_2 and M_3 to diverge. A simple measure is the ratio of correct answers above the interval midpoint to those below, or vice versa—whichever is higher. We found ratios of 3:2 or more only in two domains in the two-point condition (car prices and college ratings). Now, even such asymmetry will not produce differences in the two *M* measures unless judges' SPDFs are also skewed, and we have no way of knowing

Table 3
Hit Rates and Estimates of Interval Size Relative to Ideal "Calibra Intervals" in Experiment 2

		Interval size relative to ideal			
Domain	Hit rate[a]	M_3	M_2	Gamma simulation	Log simulation
Class enrollments	.49 (.29)	.55 (.42)	.55 (.43)	.62 (.56)	.65 (.59)
Invention dates	.60 (.16)	.55 (.24)	.59 (.26)	.71 (.25)	.66 (.26)
Walk times	.62 (.19)	.77 (.33)	.70 (.26)	.73 (.34)	.73 (.32)
July temperatures	.77 (.35)	1.33 (.72)	1.36 (.73)	1.25 (.70)	1.24 (.71)
Fertility rates	.83 (.19)	1.59 (.71)	1.15 (.31)	1.35 (.54)	1.32 (.53)

Note. Numbers in parentheses are standard deviations.
[a] Answers that exactly matched an endpoint were counted as correct.

whether that was the case here. However, given that our estimate of *M* is also lower than our simulation estimates in those domains, it is possible that our estimate of *M* is somewhat low in those cases. That implies that format differences and domain differences in that study may be even more pronounced than we report.

The degree of overconfidence was substantial in Experiment 2, 14%. However, this was considerably less than we observed in Experiment 1, and it seems to be accounted for mostly by the biasing effects of variations in interval size, especially from domain to domain. On average, individual intervals represented nearly unbiased guesses about how far apart the fractiles of the SPDF should have been if one ignores the asymmetrical effects of inward and outward errors on hit rates.

Looking across Experiments 1 and 2, one could say that the larger the number of separate estimates required, the less overconfidence was present. When a single range was requested, intervals were roughly 45% of the well-calibrated size; when two separate boundaries were requested, they were roughly 70%; and when three fractiles were requested, they were roughly 96% of the appropriate size. Later we offer further discussion of this observation. However, the comparison between Experiments 1 and 2 is not straightforward. The two studies were conducted using different domains, different information, and different populations. Therefore, Experiment 3 provides a direct test of the extent to which asking for a median estimate is indeed the reason for the relatively low overconfidence observed in Experiment 2. Finally, we note also that participants did not perform better in the domains for which they had direct, personal experience—class enrollments and walk times. Indeed, those two were among the most biased domains. Although we sampled only a few domains here, these findings suggest that overconfidence is not restricted to general knowledge, "almanac" questions.

Experiment 3

This study provides a direct comparison between the two-point and three-point methods, using a design that permits both within-participant and between-participant tests of this variable. We used two of the domains used in Experiment 2, namely July temperatures and the dates of inventions and discoveries. These domains were chosen because they were suitable for use with American participants without modification, they differed considerably from each other in the degree of overconfidence observed in Experiment 2, and they showed good convergence among different measures of bias in interval size.

Method

Participants. The participants were 42 students and staff at the University of Chicago (25 women and 17 men). One participant in each of the two conditions was replaced because repetitive or arbitrary responses indicated that they were not attending to the task. Participants were solicited via advertisements posted around campus and were paid $10.

Materials. Forty-five items were selected from each of the two domains. These included the 30 items used in Experiment 2 plus 15 new items selected randomly from the same information sources. Estimates for each item were elicited using two different formats. The *two-point* format asked for only the high and low estimates, as in the equivalent condition of Experiment 1. These were always requested in the order of lower boundary, then upper boundary. The *three-point* format asked for two boundaries and a median estimate, as in Experiment 2. Estimates for these questions were requested in one of three orders: median, lower boundary, upper boundary; lower, median, upper; and lower, upper, median.

As in Experiment 2, printed booklets were prepared with each item on its own page. Each participant received two booklets. The first booklet contained the 30 items used in Experiment 2, in shuffled order, followed by the 15 new items in that domain, also shuffled. The second booklet contained only the 30 "old" items from that domain, in a different, shuffled order.

Design. Each participant received two rounds of estimates, both from one domain. In the first round, participants received the 30 items used in Experiment 2, followed by 15 new items drawn from the same domain. In the second round, participants saw the 30 original items again. Each participant made the first 45 estimates using one format (two-point or three-point) and the last 30 using the other format. The order of formats was counterbalanced.

Three variables were manipulated between participants: domain (temperatures or discoveries), format order (two-point estimates before three-point, or vice versa), and position of the median estimate in the three-point format (first, second, or third). There were 21 participants in each domain. Within each domain, 9 participants made two-point estimates first, and 12 made three-point estimates first. In each subgroup, one third of the participants were assigned to each of the different median-estimate positions.

Procedure. The study took place in three parts, requiring about 30 min, 5 min, and 20 min, respectively. Introductory instructions indicated these time estimates, but participants were allowed to take as much or as little time as they wished. Part I consisted of the first round of estimates, using all 45 items from one of the two domains. In Part II, participants were asked to provide numerical ratings of three different types of cookies or of a recent restaurant meal, in order to provide a break and a numerical task that might reduce recall of previous estimates. These ratings were not analyzed. Part III consisted of the second round of estimates, using only the 30 items from Experiment 2, presented in a different order and in the other format.

Initial instructions described the three phases and offered instructions for whichever format was to be used in Part I. The following example is for a participant in the domain of discoveries, using the two-point format:

> For each of the 45 items in Part I, we ask you to make two estimates, by completing two statements. For example:
> In what year was [ ] invented?
> I am 90% sure that this happened after ____.
> I am 90% sure that this happened before ____.
>
> For the first statement, you should give us a year such that you believe there is a 90% chance that the discovery or invention happened <u>after</u> that. In other words, you believe there is a one in ten chance that the correct answer is earlier than this date.
>
> For the second statement, you should give us a year such that you believe there is a 90% chance that the discovery or invention happened <u>before</u> that. In other words, you believe that there is a one in ten chance that the correct answer is later than this date.

For those participants with Part I items in the three-point format, the instructions included this additional estimate:

> I think it's equally likely that this happened after or before ____.

placed in the appropriate position for that participant (i.e., before, between, or after the low and high estimates). These instructions were added:

> For the third statement, you should give us a year such that you believe that the discovery or invention is about as likely to have happened after this date as before it. In other words, you believe there is an equal chance that your guess is too early or too late.

Participants were asked to consider each item in sequence, and were told they could change their answer to the current item but could not go back to prior items.

After completing Part I, the experimenter collected the participants' booklets and handed them instructions and ratings forms for rating cookies or restaurants, which took place in another room. Following that, instructions were provided for Part III. These were similar to those used for Part I, but they used the other format (three-point or two-point). The Part III instructions included the following:

> In this final part of the study, we would like for you to revisit some of the questions that you answered in Part I. . . . **Your answers should reflect your current beliefs about each question. Do not worry about how your answers compare to what you might have said the first time you encountered these questions.**

Results

Two types of analyses were possible given the design. In one approach, we performed ANOVAs using only the 45 items presented in Part 1, ignoring the second round of estimates. Domain (temperatures or discoveries), format (two-point or three-point), and gender are all between-participants variables. In another approach, we used data from both rounds of estimates, looking only at the 30 items that appeared in both rounds. In those, format is a within-participants variable, and there is the additional between-participants variable of which format was presented first. Very similar results were obtained with both analytical methods. We present the results from the between-participants analyses, and we note the few qualitative differences found in the alternative analysis.

Hit rate. Overall, judges' 80% intervals contained the correct answer 55% of the time. Intervals in the three-point format contained the correct answers more often than in the two-point format, $F(1, 34) = 4.7$, $p = .04$, and the two domains differed, $F(1, 34) = 41.9$, $p < .001$ (see Table 4). Female judges had a higher hit rate, $F(1, 34) = 5.0$, $p = .03$. There was also a significant interaction between domain and format, $F(1, 34) = 4.4$, $p = .04$. Format made a difference in the temperatures domain (with a hit rate of .60 in the two-point format and .76 in the three-point format) but virtually no difference in the discoveries domain (.42 for both).

Results from within-participants analyses differed in two respects. The Domain × Format interaction was not significant ($F < 1$), but there was a Domain × Gender interaction, $F(1, 34) = 5.4$, $p = .03$. Women and men differed considerably in the temperatures domain (with hit rates of .81 and .60, respectively), but differed little in the discoveries domain (.51 and .48). A qualitatively similar interaction appeared in the between-participants analysis, but it was not statistically significant ($p = .11$).

MEAD and MAD. *M*, the ratio of MEAD to MAD, was higher in the three-point format than in the two-point format, $F(1, 34) = 4.2$, $p = .05$, and varied by domain, $F(1, 34) = 37.6$, $p < .001$ (see Table 4). Men were more biased than were women, $F(1, 34) = 4.2$, $p = .05$. There was a significant interaction between domain and gender, $F(1, 34) = 4.2$, $p = .05$. Women had more appropriate interval sizes than did men in the temperatures domain ($M = .93$ vs. $M = .65$), but women and men were virtually the same in the discoveries domain (both .40). Estimates obtained from simulations showed results very similar to those found with *M*, except that they also showed a significant Domain × Format interaction, $F(1, 34) = 4.7$, $p = .04$, and $F(1, 34) = 6.8$, $p = .01$, for gamma and log models, respectively. The pattern of this interaction parallels the results we reported earlier for hit rates.

The advantage of the three-point format seems to derive from both smaller errors and larger intervals. MAD was about 10% lower in the three-point format, and intervals were about 13% larger, although neither of those components was itself statistically significant. The within-participants analysis looks a little different. There the three-point format showed wider intervals, $F(1, 34) = 8.1$, $p < .01$, but no significant advantage in MAD ($F < 1$).

The effect of gender seemed to result more from differences in interval width than in accuracy. Women's intervals were almost 20% wider than men's, and their MADs were about 6% lower. However, neither of these differences was statistically reliable.

Order of estimates. In the three-point format, we always asked for the lower bound before the upper, but we varied the position in

Table 4
Hit Rates and Estimates of Interval Size Relative to Ideal "Calibra Intervals" in Experiment 3

Independent variable	Hit rate[a]	Interval size relative to ideal: *M*	Gamma simulation	Log simulation
Format				
Two-point	.51 (.17)	.53 (.25)	.58 (.23)	.55 (.24)
Three-point	.59 (.20)	.67 (.36)	.73 (.31)	.68 (.33)
Domain				
Temperatures	.68 (.17)	.82 (.31)	.85 (.27)	.83 (.27)
Discoveries	.42 (.11)	.41 (.16)	.48 (.15)	.42 (.14)
Gender				
Women	.60 (.20)	.69 (.37)	.73 (.32)	.70 (.33)
Men	.51 (.18)	.53 (.24)	.61 (.24)	.55 (.25)

Note. The number of male and female participants differed across format by domain cells. To best represent format and domain means independent of gender, we differentially weighted cases to equate the weight of men and women in each cell. Numbers in parentheses are standard deviations.
[a] Answers that exactly matched an endpoint were counted as correct.

which we asked for the median estimate. We looked at the effects of order by examining only those questions asked in the three-point format, using the 30 questions that participants saw twice. Every participant answered those questions in the three-point format, either in the first round or the second.

We performed between-participants ANOVAs with four independent variables: estimate order, domain, gender, and whether the three-point questions were received before or after having given two-point responses. The dependent variables were the same ones used in the other analyses. No main effects of estimate order approached significance. Hit rate and, marginally, *M* showed an Order × Domain interaction. Those measures and MAD also showed an Order × Gender interaction. In the temperatures domain, participants appeared to do better when the median came first or second. In the discoveries domain, they did better when it came second or third. Women did better when the median was second or third; men did better when it was first or second. Clearly, if order of estimates has effects, they are complex ones.

Very little of the observed overconfidence in Experiment 3 is attributable to variability. Using the methods described earlier, we estimated that contribution of variability to overconfidence was 0 in the two-point format and .03 in the three-point format.

Discussion

The results of the present study confirm that asking for a median estimate in addition to boundaries does make a difference. Also, the results with the three-point format are in line with those observed in Experiment 2. (Although it appears that there was less overconfidence in Experiment 3, when we compared only those items that the two studies had in common, there was little overall difference.) Within Experiment 3, we find that adding the median estimate improved hit rates considerably, although all of the improvement was in one domain. We did not replicate the finding in Experiment 2 that providing estimates in the order of low–median–high produced more overconfidence. Indeed, we found no clear effect of where we put the median estimate in the sequence.

In sum, the results of Experiment 3 support the hypothesis that making additional, separate estimates about different parts of one's subjective probability distribution reduces overconfidence. Doing so seems both to improve the accuracy of one's estimates and to increase the size of one's intervals. At the same time, we continue to find noticeable differences between populations and between domains. These differences suggest caution in generalizing from results, but they may also provide interesting clues to underlying processes.

General Discussion

People are grossly overconfident when they provide subjective intervals for a quantitative estimate. Findings with two-choice questions are much different; some domains exhibit overconfidence and others underconfidence, but the net effect is small when representative sets of questions are sampled from multiple domains. Some forms of interval estimates, in contrast, show overconfidence of 40% or more even with representative sets of questions (see also Klayman et al., 1999).

Taking a lesson from research on two-choice questions, we examined the possibility that overconfidence in intervals might be due to effects of variability in interval widths. Making an interval 10 years or 6° or 1,700 m too narrow has more effect on hit rates than do the opposite errors. This effect causes a bias toward overconfidence if there is variability from item to item for a given person, from person to person in a given domain, and from domain to domain. It is reasonable to imagine that judges do not realize or consider this statistical quirk and offer intervals that would otherwise be appropriately sized on average. We developed several different methods with which to analyze interval width independently of hit rate. All methods agree: Variability contributes to overconfidence, but the main cause is that subjective intervals are too narrow. Nevertheless, it is important for researchers to keep variability in mind, because the amount of variability in items is partially a function of which items are chosen. For example, in Experiment 2, we used domains that vary greatly in the extent to which intervals are overly narrow and for which variability accounted for half of the total observed overconfidence in that study.

We also find that the format by which subjective intervals are solicited has a large effect on overconfidence and interval size. For example, one can ask for a range that the judge is 90% sure contains the correct answer, or ask separately for a point with 5% chance of being too low and another with 5% chance of being too high, or ask for those two estimates plus a median estimate. The first will produce the most overconfidence; the last will produce the least.

Having established that interval estimates are too narrow, we would of course like to understand when and why. We are not yet in a position to answer those questions definitively. However, the pattern of results in the present experiments offers some interesting suggestions. One obvious possibility is that narrow intervals are the result of anchoring (presumably on a best-guess point estimate) and insufficient adjustment. To some extent, though, this merely restates the phenomenon. A number of investigators have suggested that insufficient adjustment itself reflects confirmation biases and similar cognitive processes (Chapman & Johnson, 2002). Furthermore, one would still want to know why adjustment was more insufficient in some formats than in others. At the same time, some of our results seem not to fit an anchoring story very easily. It is hard to know just where a judge's anchor lies, but it is likely to be near the median (i.e., where the answer is believed to be equally likely to be above or below). Making the median estimate explicit should make the anchor even more salient or, perhaps, have no effect (because judges already start from their best guess). Instead, asking for an explicit median increases the size of subjective intervals.

What, then, are the likely sources of bias? When judges collect information, either from their heads or from the outside world, they obtain only a limited sample of the potential information. Moreover, the sample they draw is probably not unbiased. Because of the nature of human associative memory, information that has many semantic connections with already retrieved information will have a retrieval advantage, making the sample more consistent than a random sample of evidence would be. Confirmation biases (see Klayman, 1995) will further contribute to the excessive consistency of information and interpretations of its implications for the answer. People fully anticipate neither the consequences of confirmation biases (Klayman, 1995) nor of limited and biased samples more generally (Fiedler, 2000). They treat their sample of

evidence too much as though it were a reliable and unbiased representation of the world.

The above explanation seems like a throwback to similar cognitive explanations offered over 20 years ago for overconfidence in two-choice questions (e.g., by Koriat et al., 1980). Subsequent research has raised serious doubts about the appropriateness of such explanations for two-choice questions. So why revive them for intervals? Most fundamentally, there seems to be fairly little overall bias toward overconfidence across domains in two-choice questions but a strong overall bias in intervals. Klayman et al. (1999) argued that confirmation biases are expected to have more impact on the latter. Who has a longer average life expectancy, Argentines or Canadians? The question offers two explicit alternatives: Argentines live longer, or Canadians do. Thus, both possibilities are considered more or less equally, producing error-prone, but not heavily biased, judgments of the balance of evidence. In contrast, when asked for a single range estimate, the judge starts with some initial impression of, say, the longevity of Canadians and must then judge the appropriate precision of that impression. This is more akin to the situation in which the validity of a single hypothesis is being evaluated, which is when confirmation biases have the most effect (Klayman, 1995; McKenzie, 1999).

What about the format differences observed in the present studies? When judges are asked for separate low and high 90% judgments rather than a single 80% range, their intervals become wider, and the centers of their intervals are closer to the correct answers. An evidence-sampling explanation fits this case. When asked to provide a lower bound, the judge is implicitly asked to consider what the lowest plausible answer could be (specifically, one that is 90% sure to be too low). This elicits a search for evidence that Canadians are prone to early death (e.g., because of the hardships of harsh weather conditions). The separate request for the highest plausible answer elicits a search for evidence that Canadians are hardy (e.g., because of the rigors of harsh weather conditions). The result is two separate searches for evidence in two different regions of knowledge. This increases the range of evidence considered (making judgments more accurate) while making the variability of implications more apparent (increasing interval width). Now another request is added, namely for the point at which the correct answer seems to be equally likely to be above or below. This might be seen as an incitement to anchor, but in the context of the two other required estimates, it provokes a third search in yet another region of knowledge, again increasing both accuracy and the sense of the variety of implications of different evidence.

Our explanation of overconfidence in interval judgments must be regarded as speculative, pending further studies of the cognitive processes underlying confidence judgments. Still, the differences between interval and two-choice questions and among different interval-judgment formats is consistent with an explanation in terms of limitations and biases in gathering and interpreting evidence.

Having appropriate confidence is important for making appropriate risky decisions, for knowing when to seek advice and information, and for communicating one's knowledge. Judging appropriate confidence is not easy. Furthermore, there is a general tendency toward overconfidence, which can be quite severe in some conditions but is not universal. Clearly, there is much more that remains to be learned about how people form subjective judgments of confidence. We are optimistic that future research into differences between formats, people, and domains will advance our understanding of the psychology of confidence and our ability to help people be better judges of confidence. In the meantime, we can say that if you want to obtain a well-calibrated estimate, ask for a comparison between two options, or else ask separately about different parts of the range of possible answers. Also, be careful what you ask about, and ask a woman.

References

Alpert, M., & Raiffa, H. (1982). A progress report on the training of probability assessors. In D. Kahneman, P. Slovic, & A. Tversky (Eds.), *Judgment under uncertainty: Heuristics and biases* (pp. 294–305). Cambridge, England: Cambridge University Press.

Ayton, P., & McClelland, A. G. R. (1997). How real is overconfidence? *Journal of Behavioral Decision Making, 10,* 279–285.

Barber, B. M., & Odean, T. (2001). Boys will be boys: Gender, overconfidence, and common stock investments. *Quarterly Journal of Economics, 116,* 261–292.

Block, R. A., & Harper, D. R. (1991). Overconfidence in estimation: Testing the anchoring-and-adjustment hypothesis. *Organizational Behavior and Human Decision Processes, 49,* 188–207.

Brenner, L. (2000). Should observed overconfidence be dismissed as a statistical artifact? Critique of Erev, Wallsten, and Budescu (1994). *Psychological Review, 107,* 943–946.

Brenner, L. A. (2003). A random support model of the calibration of subjective probabilities. *Organizational Behavior and Human Decision Processes, 90,* 87–110.

Budescu, D. V., Erev, I., & Wallsten, T. S. (1997). On the importance of random error in the study of probability judgment: Part I. New theoretical developments. *Journal of Behavioral Decision Making, 10,* 157–171.

Chapman, G. B., & Johnson, E. J. (2002). Incorporating the irrelevant: Anchors in judgments of belief and value. In T. Gilovich, D. Griffin, & D. Kahneman (Eds.), *Heuristics and Biases: The Psychology of Intuitive Judgment* (pp. 120–138). Cambridge, England: Cambridge University Press.

Clemen, R. T. (1996). *Making hard decisions: An introduction to decision analysis* (2nd ed.). Boston: PWS-Kent Publishing.

Clemen, R. T. (2001). Assessing 10-50-90s: A surprise. *Decision Analysis Newsletter, 20*(1), 2, 15.

Dawes, R. M., & Mulford, M. (1996). The false consensus effect and overconfidence: Flaws in judgment or flaws in how we study judgment? *Organizational Behavior and Human Decision Processes, 65,* 201–211.

Erev, I., Wallsten, T. S., & Budescu, D. V. (1994). Simultaneous over- and underconfidence: The role of error in judgment processes. *Psychological Review, 101,* 519–527.

Fiedler, K. (2000). Beware of samples! A cognitive-ecological sampling approach to judgment biases. *Psychological Review, 107,* 659–676.

Gigerenzer, G., Hoffrage, U., & Kleinbölting, H. (1991). Probabilistic mental models: A Brunswikian theory of confidence. *Psychological Review, 98,* 506–528.

Griffin, D., & Tversky, A. (1992). The weighing of evidence and the determinants of confidence. *Cognitive Psychology, 24,* 411–435.

Hoch, S. J. (1985). Counterfactual reasoning and accuracy in predicting personal events. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 11,* 719–731.

Hogarth, R. M. (1978). A note on aggregating opinions. *Organizational Behavior and Human Performance, 21,* 40–46.

Johnson, N. L., Kotz, S., & Balakrishnan, N. (1995). *Continuous univariate distributions* (Vol. 2). Chichester, England: Wiley.

Juslin, P. (1994). The overconfidence phenomenon as a consequence of

informal experimenter-guided selection of almanac items. *Organizational Behavior and Human Decision Processes, 57,* 226–246.

Juslin, P., & Olsson, H. (1997). Thurstonian and Brunswikian origins of uncertainty in judgment: A sampling model of confidence in sensory discrimination. *Psychological Review, 104,* 344–366.

Juslin, P., Olsson, H., & Björkman, M. (1997). Brunswikian and Thurstonian origins of bias in probability assessment: On the interpretation of stochastic components of judgment. *Journal of Behavioral Decision Making, 10,* 189–209.

Juslin, P., Wennerholm, P., & Olsson, H. (1999). Format dependence in subjective probability calibration. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 25,* 1038–1052.

Juslin, P., Winman, A., & Olsson, H. (2000). Naive empiricism and dogmatism in confidence research: A critical examination of the hard–easy effect. *Psychological Review, 2,* 384–396.

Kirk, R. E. (1995). *Experimental design: Procedures for the behavioral sciences* (3rd ed.). Pacific Grove, CA: Brooks/Cole.

Klayman, J. (1995). Varieties of confirmation bias. In J. R. Busemeyer, R. Hastie, & D. L. Medin (Eds.), *Decision making from the perspective of cognitive psychology* (pp. 385–418). New York: Academic Press.

Klayman, J., & Burt, R. S. (1998). *Individual differences in confidence and experiences in social networks.* Unpublished manuscript, Center for Decision Research, Graduate School of Business, University of Chicago.

Klayman, J., Soll, J. B., González-Vallejo, C., & Barlas, S. (1999). Overconfidence: It depends on how, what, and whom you ask. *Organizational Behavior and Human Decision Processes, 79,* 216–247.

Koehler, D. J., Brenner, L. A., & Griffin, D. (2002). The calibration of expert judgment: Heuristics and biases beyond the laboratory. In T. Gilovich, D. Griffin, & D. Kahneman (Eds.), *Heuristics and biases: The psychology of intuitive judgment* (pp. 686–715). Cambridge, England: Cambridge University Press.

Koriat, A., Lichtenstein, S., & Fischhoff, B. (1980). Reasons for confidence. *Journal of Experimental Psychology: Human Learning and Memory, 6,* 107–118.

Lichtenstein, S., & Fischhoff, B. (1981). *The effects of gender and instructions on calibration* (Tech. Rep. PTR-1092-81-7). Eugene, OR: Decision Research.

Lichtenstein, S., Fischhoff, B., & Phillips, L. D. (1982). Calibration of probabilities: The state of the art to 1980. In D. Kahneman, P. Slovic, & A. Tversky (Eds.), *Judgment under uncertainty: Heuristics and biases* (pp. 306–334). Cambridge, England: Cambridge University Press.

Lundeberg, M. A., Fox, P. W., & Punćochaŕ, J. (1994). Highly confident but wrong: Gender differences and similarities in confidence judgments. *Journal of Educational Psychology, 86,* 114–121.

McKenzie, C. R. M. (1999). (Non)Complementary updating of belief in two hypotheses. *Memory & Cognition, 27,* 152–165.

McKenzie, C. R. M., & Amin, M. B. (2002). When wrong predictions provide more support than right ones. *Psychonomic Bulletin & Review, 9,* 821–828.

O'Connor, M., & Lawrence, M. (1989). An examination of the accuracy of judgment confidence intervals in time series forecasting. *International Journal of Forecasting, 8,* 141–155.

Pallier, G. (2003). Gender differences in the self-assessment of accuracy on cognitive tasks. *Sex Roles, 48,* 265–276.

Pfeifer, P. E. (1994). Are we overconfident in the belief that probability forecasters are overconfident? *Organizational Behavior and Human Decision Processes, 58,* 203–213.

Russo, J. E., & Schoemaker, P. J. H. (1992). Managing overconfidence. *Sloan Management Review, 33,* 7–17.

Selvidge, J. E. (1980). Assessing the extremes of probability distributions by the fractile method. *Decision Sciences, 11,* 493–502.

Sniezek, J. A., & Buckley, T. (1991). Confidence depends on level of aggregation. *Journal of Behavioral Decision Making, 4,* 263–272.

Soll, J. B. (1996). Determinants of overconfidence and miscalibration: The roles of random error and ecological structure. *Organizational Behavior and Human Decision Processes, 65,* 117–137.

Soll, J. B. (1999). Intuitive theories of information: Beliefs about the value of redundancy. *Cognitive Psychology, 38,* 317–346.

Spetzler, C. S., & Staël von Holstein, C.-A. S. (1975). Probability encoding in decision analysis. *Management Science, 22,* 340–358.

Suantak, L., Bolger, F., & Ferrell, W. R. (1996). The hard–easy effect in subjective probability calibration. *Organizational Behavior and Human Decision Processes, 67,* 201–221.

Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. *Science, 185,* 1124–1131.

von Winterfeldt, D., & Edwards, W. (1986). *Decision analysis and behavioral research.* Cambridge, England: Cambridge University Press.

Wallsten, T. S., Erev, I., & Budescu, D. V. (2000). The importance of theory: Response to Brenner (2000). *Psychological Review, 107,* 947–949.

Winer, B. J. (1971). *Statistical principles in experimental design* (2nd ed.). New York: McGraw-Hill.

Yaniv, I., & Foster, D. P. (1995). Graininess of judgment under uncertainty: An accuracy informativeness trade-off. *Journal of Experimental Psychology: General, 124,* 424–432.

Yaniv, I., & Foster, D. P. (1997). Precision and accuracy of judgmental estimation. *Journal of Behavioral Decision Making, 10,* 21–32.

(*Appendix follows*)

Appendix

Models and Procedures Used to Estimate Interval-Size Bias on the Basis of the Observed Hit Rate and the Covariance Between Interval Size and Hits

Gamma Model

This model corresponds to Equation 1 in the text, $W_i = xW_i^*$, where W_i is the width of the interval the judge gives for estimate i and W_i^* is the Calibra size. Across a set of items, variable x has a mean of $\bar{x}$ and some mean-zero variation, v_i, around that, so

$$W_i = W_i^*(\bar{x} + v_i). \tag{A1}$$

We define a binary variable H_i, such that $H_i = 1$ if the interval for item i contains the correct answer (a hit) and $H_i = 0$ otherwise. By definition, for p% intervals W_i^* varies from item to item, being wider when knowledge is low and narrower when knowledge is high, so that the probability of a hit is always p. So $\text{cov}(W_i^*, H_i) = 0$. Because $\bar{x}$ is a constant, $\text{cov}(\bar{x}W_i^*, H_i) = 0$ also. That is, if there were no variation (no v_i), there would be no covariance between the width of the interval and whether or not there is a hit.

In the presence of this variation, however, the probability of a hit does vary with interval size. A larger v_i means a larger interval for the same degree of knowledge and, thus, a higher chance of a hit. On average, wider intervals will be associated with higher hit rates, that is, $\text{cov}(W_i, H_i) > 0$. By observing the size of this covariance in the data, we can get a clue as to the amount of bias and variation across a given set of items (typically, for a given participant in a particular domain).

We need, therefore, to find the mathematical relationship between the observable quantity $\text{cov}(W_i, H_i)$ and the theoretical quantities of interest, $\bar{x}$ and v_i. Beginning with Equation A1, we have:

$$W_i = \bar{x}W_i^* + v_iW_i^*$$

$$\text{cov}(W_i, H_i) = \text{cov}(\bar{x}W_i^*, H_i) + \text{cov}(v_iW_i^*, H_i). \tag{A2}$$

To proceed, we need to make an additional, simplifying assumption, namely that H_i is a function only of $\bar{x}$ and v_i and not of W_i^*. This implies that W_i^* is independent of v_i as well as H_i and any other functions of v_i. Substantively, we are assuming that the probability of a hit does not depend on the level of knowledge that a judge has, at least within a given domain. Early work on overconfidence suggested that people were more overconfident when they knew less. However, recent research has shown that this hard–easy effect is mainly an artifact of how the hard and easy questions are selected (e.g., Dawes & Mulford, 1996; Gigerenzer et al., 1991; Juslin et al., 2000; Klayman et al., 1999). Thus, the assumption that error is unrelated to well-calibrated interval width is reasonable. Given this independence assumption, plus the fact that $\bar{x}$ is a constant, Equation A2 simplifies to

$$\text{cov}(W_i, H_i) = \text{cov}(v_iW_i^*, H_i).$$

The next two steps are based on a standard definition of covariance, $\text{cov}(X, Y) = \text{E}(XY) - \text{E}(X)\text{E}(Y)$, or equivalently, $\text{E}(XY) = \text{E}(X)\text{E}(Y) + \text{cov}(X, Y)$. Continuing, then,

$$\begin{aligned}\text{cov}(W_i, H_i) &= \text{cov}(v_iW_i^*, H_i) = \text{E}(v_iW_i^*H_i) - \text{E}(H_i)\,\text{E}(v_iW_i^*)\\ &= \text{E}(W_i^*)\,\text{E}(v_iH_i) + \text{cov}(W_i^*, v_iH_i)\\ &\quad - \text{E}(H_i)\{\text{E}(v_i)\,\text{E}(W_i^*) + \text{cov}(W_i^*, v_i)\}.\end{aligned} \tag{A3}$$

By the definition of v_i, $\text{E}(v_i) = 0$, and the independence assumption implies that $\text{cov}(W_i^*, v_i) = 0$ and $\text{cov}(W_i^*, v_iH_i) = 0$, so Equation A3 simplifies to

$$\begin{aligned}\text{cov}(W_i, H_i) &= \text{E}(W_i^*)\,\text{E}(H_iv_i)\\ &= \text{E}(W_i^*)\{\text{E}(H_i)\,\text{E}(v_i) + \text{cov}(H_i, v_i)\}\\ &= \text{E}(W_i^*)\text{cov}(H_i, v_i)\\ &\qquad [\text{because } \text{E}(v_i)=0].\end{aligned} \tag{A4}$$

From Equation A1, it is easy to show that $\text{E}(W_i) = \bar{x}\text{E}(W_i^*)$. Therefore, we can rewrite Equation A2 as

$$\frac{\text{cov}(W_i, H_i)}{E(W_i)} = \frac{\text{cov}(H_i, v_i)}{\bar{x}}. \tag{A5}$$

The quantities on the left side of Equation A5 can be estimated from observable behaviors. The numerator is simply the covariance between interval width and the binary variable that indicates hits and misses. The denominator is estimated by mean interval width.

We used Monte Carlo simulations to produce a variety of distributions of interval sizes, using a variety of different gamma functions for $(\bar{x} + v_i)$. Bounded at zero, gamma distributions have two parameters, shape and scale, that uniquely determine the mean $(\bar{x})$ and the standard deviation around it, σ_v. We then scored simulated hits and misses by matching each simulated interval to a standard normal distribution of correct answers. Luckily, different combinations of $\bar{x}$ and σ_v produce unique combinations of hit rate and covariance that can be matched to observed results.

Log Model

The logic for our alternative model is the same, but starting from a different assumption about how actual and ideal interval sizes are related, as shown in Equation 2 of the text, namely $\log(W_i/W_i^*) = x$. Again, we partition z into its mean, $\bar{x}$, and mean-zero variation around it, v_i. So,

$$\log(W_i/W_i^*) = \bar{x} + v_i$$

$$\log W_i = \log W_i^* + \bar{x} + v_i$$

$$\begin{aligned}\text{cov}(\log W_i, H_i) &= \text{cov}([\log W_i^* + \bar{x} + v_i], H_i)\\ &= \text{cov}(\log W_i^*, H_i) + \text{cov}(\bar{x}, H_i) + \text{cov}(v_i, H_i).\end{aligned}$$

As we discussed earlier, $\text{cov}(W_i^*, H_i) = 0$, so logically $\text{cov}(\log W_i^*, H_i) = 0$, too; $\text{cov}(\bar{x}, H_i) = 0$ because $\bar{x}$ is a constant. Thus,

$$\text{cov}(\log W_i, H_i) = \text{cov}(v_i, H_i). \tag{A6}$$

This time, our Monte Carlo simulations used normal distributions for $(\bar{x} + v_i)$, with different means and standard deviations. We then matched the resulting distribution of intervals to a standard normal distribution of correct answers to calculate hits and misses and the covariance. Again, different combinations of $\bar{x}$ and σ_v produce unique combinations of hit rate and covariance that can be matched to the observed hit rate and the observed covariance between log-interval widths and hits.

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